(24P-0619) (BCS-3D) (Musawar Ali Shah) 1

① Even - Even :

$$R = (aa + bb + (ab+ba)(aa+bb)^*(ab+ba))^*$$

② Even - Odd :-

~~$R \Rightarrow (b^* + \epsilon)(aa + bb + (ab+ba)(ab+ba)$~~

~~$R = (b^* + \epsilon)($~~

$$R = b^* (aa + bb + (ab+ba)(aa+bb)^*(ab+ba))^* (bb)^* b$$

③ Odd - Even :

$$R \Rightarrow a^* (aa + bb + (ab+ba)(aa+bb)^*(ab+ba))^* (aa)^* a$$

④ Odd - Odd :-

$$R = (aa)^* a (aa + bb + (ab+ba)(aa+bb)^*(ab+ba))^* (bb)^* b$$

$$+$$

$$(bb)^* b \ (aa + bb + (ab+ba)(aa+bb)^*(ab+ba))^* (aa)^* a$$

5.) String that begins & ends with same latter :

$$R = a(a+b)^* a + b(a+b)^* b$$

6.) String that begins with 'a' & contains atleast two b's :

$$R \Rightarrow a(a+b)^* b(a+b)^* b(a+b)^*$$

7.) String that ends with 'ab' and contains no substrings 'bb'

$$R = a^* (b + \epsilon)(a + ba)^* ab$$

8.) contains 'ab' as substring but donot contains 'ba' as substring :

$$R = a^* b^*$$

9.) Contains both ab and ba as substrings: (in any order). (2)

$R = (a+b)^* ab (a+b)^* ba (a+b)^*$

10.) Start with aa or end with bb (or both).

$R = aa(a+b)^* (aa+bb)$

11.) No occurance of substring aaa.

~~$(\epsilon + a + aa)(b+ab+aab)^*$~~

$R = (\epsilon + a + aa)(b+ba+baa)^*$

12.) No occurance of substring 'bbb'.

$(\epsilon + b + bb)(a+ab+abb)^*$

13.) String with no adjacent equal symbols (i.e. strictly alternating symbols):

$R \Rightarrow ((ab+ba)^* + a+b)$

14.) do not start with b and donot contain 'bbb'

$R \Rightarrow (a+ab+ba+a^*ba^* + a^*bba^*)$

15.) avoid aba as a substring:

$R = (a^* + b^* + b^*a^*ab + abb^*a^*)$

16.) with exactly 'aaa':

$R = (b^* ab^* ab^* ab^*)$

17.) with atmost two b's:

$R = (a^* + a^*ba^* + a^*ba^*ba^*)$

18.) Atleast three a's & atmost one b.

$R = (aaaa^* + a^*baaaa^* + a^*abaaa^* + a^*aabaa^* + a^*aaaba^*)$

19) Exactly one occurance of the substring ab. ③

$R = (b^*a^*ab)(b^*a^*)^*$

20) Exactly two occurance of substring ba (occurance may overlap)

$R = (a^*b^* \, ba \, a^*b^* \, ab \, a^*b^*)$

21) String in which the number of a's is a multiple of 3 (no restriction on b)

~~$(b^*aaa)(b^*a$~~

$R = (b^*aaab^*)(b^*ab^*ab^*)^*$

22) Strings in which the number of b's is congruent to 1 (mod 3)

$R = a^*ba^*(ba^*ba^*ba^*)^*$

23) Strings whose length is even.

$R = ((a+b)(a+b))^*$

24) Odd-length

$R = (a+b)((a+b)(a+b))^*$

25) String where the total number of a's is congruent to 2 (mod 4).

$R = (b^*ab^*ab^*ab^*ab^*)^* \, ba^*b^*ab^*$

26) Strings that start with b and contain an even number of a's:

$R = (b(aa)^* + bb^*(ab^*ab^*)^*)$

27.) String that end with a and contain an odd number of b's

$$R = (a^* ba + (ba^* ba))$$

28.) String that string and end with a and have an number of b's divisible by 2:

$$R = aa^* (ba^* ba^*)^* a + a^* a (ba^* ba^*) a$$

29.) String that start with ab and have a total length divible by 3.

$$ab ((a+b)(a+b)(a+b))^*$$

30.) end with ba and have a number of a's divisible by 3.

$$R \Rightarrow b^* a b^* a b^* (ab^* ab^* ab^*)^* ba$$

31.) Whose third symbol from the right (if it exists) is 'a'.

$$R = (a+b)^* a (a+b)(a+b)$$

32.) String whose second symbol is b (strings of length < 2 are excluded)

$$R = ((a+b)b (a+b)^*)$$

33.) Strings that end with a and contain b only in blocks of even length (i.e b's appear in pairs).

$$(a + bb)^* a$$

34.) Strings in which every block of consecutive a's has length at most 2.

$$R = (b + ab + ba + baa + aab + b^* aa b^*)$$

35.) Strings in which every block of consecutive b's has length exactly 2.

$R = a^* bb a^*$

36.) Strings that contain aa but donot contain bb.

$R = ((a+ba)^* aa(a+ba)^* + (a+ab)^* aa(a+ab)^*)$

37.) Strings that contain bb but donot contain aa.

$R \Rightarrow ((b+ab)^* bb (b+ab)^*) + ((b+ba)^* bb (b+ba)^*)$

38.) String that contain aa exactly once.

$R = ((ab)^* aa (ab)^* + (ba)^* aa (ba)^*) + (b^* aa b^*)$

39.) Strings that contain bb exactly once.

$R \Rightarrow ((ab)^* bb (ab)^*) + ((ba)^* bb (ba)^*) + (a^* bb a^*)$

40.) Strings that contain exactly one of the substrings ab or ba (but not both).

$R = ((aa^* bb^*) + (bb^* aa^*)$

41.) Strings in which every a is immediately followed by b (i.e. no a is followed by a or end of string).

$R = (b+ab)^*$

42.) Strings in which every b is immediately followed by a.

$R \Rightarrow (a+ba)^*$

43.) Strings where every a is both proceeded & followed by b (i.e. all a occur inside bab patterns):

$R = (b+bab)^*$

44.) Strings where all occurrences of ab are immediately followed by b (i.e. only bbb may realize ab).

$R \Rightarrow (a^* + b^*)^* + (abb)^* + (a^* + b^*)^*$

45. String where ~~all~~ any b that appears must be part of a block baa (i.e. b only occurs as the first symbol of baa):

$a^* (baa) a^*$

46.) Strings with an even number of a's that donot contain bb.

$R = ((a+ba)(a+ba))^*$

47.) Strings with an odd number of b's that begin with a and end with a

$R = a(a^* b a^* (ba^* ba^*)^*) a$

50.) ~~Consider~~ Write first 10 strings (words) generated by this expression

$L = \{\varepsilon, ab, ba, aab, aba, abb, bab, bba, aab\}$

48.) Regular expression for your name:

(MUHAMMAD)(MUSAWAR)(ALI)(SHAH)

49.) Regular expression for a valid email address:

$R = ((a-z)(A-Z)(0-9)(@gmail.com)$